Practice · Arithmetic

Each question lists its options; the right one is marked **Right answer**. *Feedback* is what students see after they answer. In a typed question the blank is in square brackets, with the accepted answers separated by `|`.

(mc) L1 · The multiply sign

Which symbol multiplies in C?

- `x`
- `×`
- **Right answer:** `*`
- `^`

Feedback: C uses the star `*` for multiplication. `x` would be a variable name, and `^` is not "to the power of" in C.

(mc) L1 · The remainder sign

Which symbol gives the remainder after division?

- `/`
- `\`
- `rem`
- **Right answer:** `%`

Feedback: `%` is called modulo. `7 % 3` is 1, because 3 goes into 7 twice and 1 is left over.

(mc) L1 · Whole number division

What is `7 / 2` in C, when both numbers are `int`?

- `3.5`
- **Right answer:** `3`
- `4`
- `1`

Feedback: Dividing an `int` by an `int` gives an `int`. The .5 is thrown away, not rounded. So 3.

(mc) L2 · Predict the four lines

With `int a = 7;` and `int b = 3;`, what do these four lines print, in order?

```
printf("%d\n", a + b);
printf("%d\n", a / b);
printf("%d\n", a % b);
printf("%d\n", 2 + 3 * 4);
```

- 10, 2.33, 1, 20
- 10, 2, 3, 14
- **Right answer:** 10, 2, 1, 14
- 10, 3, 1, 20

Feedback: 7 + 3 = 10. 7 / 3 = 2 (the decimal part is thrown away). 7 % 3 = 1 (the remainder). 2 + 3 × 4 = 14, because multiplication comes first.

(mc) L2 · Getting the decimals

Which expression gives `3.5`?

- `7 / 2`
- **Right answer:** `7.0 / 2`
- `7 % 2`
- `(int) 7 / 2`

Feedback: If at least one side is a `double`, C keeps the decimals. `7.0 / 2` is 3.5. `7 / 2` is 3.

(mc) L2 · Brackets

What is `(2 + 3) * 4`?

- `14`
- `24`
- **Right answer:** `20`
- `11`

Feedback: Brackets first: 2 + 3 = 5. Then 5 × 4 = 20. Without brackets, `2 + 3 * 4` is 14.

(fib) L3 · Remainder

What number does `printf("%d\n", 10 % 3);` print? [1]

Feedback: 3 goes into 10 three times (9), and 1 is left over.

(fib) L3 · Integer division

What number does `printf("%d\n", 9 / 2);` print? [4]

Feedback: 9 / 2 is 4.5, and the .5 is thrown away. So 4.

(fib) L3 · Is it even

Complete the expression that is 0 when `n` is even and 1 when `n` is odd: n [% 2|%2]

Feedback: `n % 2`. The remainder after dividing by 2 is 0 for even numbers and 1 for odd numbers.